

Snake Game – Documentation

title

1. Introduction

This project is a simple Snake Game developed using C# and Windows Forms. The player controls a snake using arrow keys. The snake grows when it eats food, and the game ends upon collision with walls or the snake's own body. The project uses a timer-based game loop and GDI+ drawing inside a Panel control.

2. Tools & Technologies Used

- C#
- .NET Windows Forms
- Visual Studio
- Timer-based game loop
- Event-driven programming
- GDI+ Graphics

3. Features

- Smooth snake movement
- Arrow key controls
- Random food generation
- Score and high score tracking
- Self and wall collision detection
- Start, Pause, Resume functionality
- Dynamic drawing via Panel Paint event

4. How the Game Works

1. Start Game initializes snake, food, direction, and score.
2. Timer Tick moves the snake forward.
3. Keyboard input changes direction.
4. Collision detection checks for game over.
5. Graphics are drawn using the Paint event.

5. Code Structure

Main files:

- Form1.cs: Game logic, movement, collision, scoring

- Form1.Designer.cs: UI layout
- Program.cs: Application entry
- SnakeGame.csproj: Project settings

6. Conclusion

This project demonstrates skills in event-driven programming, handling user input, using timers, drawing graphics, and implementing game logic. The game fulfills all assignment requirements and provides a complete interactive experience.